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Next Gen-AI Enabled New Type of Power System—Article

AI-Optimized Decision-Making in Energy Systems: Toward a Decision-Aware Machine Learning Framework

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ABSTRACT

Machine learning (ML) and mathematical programming (MP) models are essential for predicting uncertain system parameters and optimizing decision-making in energy systems. However, incorporating the impact of prediction errors into ML models to inform MP models remains a significant challenge. To address this issue, we propose an artificial intelligence (AI)-native optimization (AIOpti) system that enables AI to be aware of the impact of predictions on decision-making through a deep fusion of ML and MP models. This AI-native deep fusion distinguishes AIOpti from existing research, as the latter separates accuracy-oriented learning from problem-oriented learning. When applied to scenarios such as virtual power plant operations and distributed energy management based on real-world data, AIOpti achieves lower computational costs, improved convergence, and robust performance in complex systems, enabling ML models to effectively account for the impact of prediction errors on MP models. Furthermore, our enhanced AIOpti system not only maximizes predictive accuracy but also improves decision-making quality.

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1. Introduction

High penetration of clean energy generation is crucial for decarbonizing the energy sector, especially given the pressing need to mitigate climate change [1,2]. Clean energy sources such as wind and solar power hold tremendous potential, but their variable outputs and geographical dispersion present complex challenges for integration into urban energy systems [3,4]. This is where predictive machine learning (ML) models and mathematical programming (MP) techniques can play a transformative role in addressing real-world energy problems [5–7]. ML models can increase electricity market efficiency by accurately predicting electricity prices and energy supply-and-demand patterns, thereby improving the integration of renewable resources [8]. Following the prediction and estimation of system uncertainties, MP models—particularly mixed-integer linear programming (MILP) models—are commonly used to optimize and analyze energy system challenges, including electricity market decarbonization [9], wind energy integration [10], and renewable investments [11]. For instance, dynamic pricing in day-ahead markets incentivizes distributed suppliers to preschedule power generation and encourages consumers to adjust their demand patterns in response to price signals [12,13]. Traditional decision-making in energy systems often employs a two-stage approach: System parameters are predicted first, followed by the optimization of strategic decisions based on these forecasts [14,15]. However, as shown in Fig. 1(a), higher prediction accuracy does not always contribute to improved decision-making; in fact, in some cases, it can even have adverse effects. Conventional ML approaches primarily focus on minimizing prediction errors, often overlooking the broader implications of prediction accuracy on downstream optimization processes [16]. Consequently, conducting ML and MP modeling separately for engineering problems can lead to inaccurate

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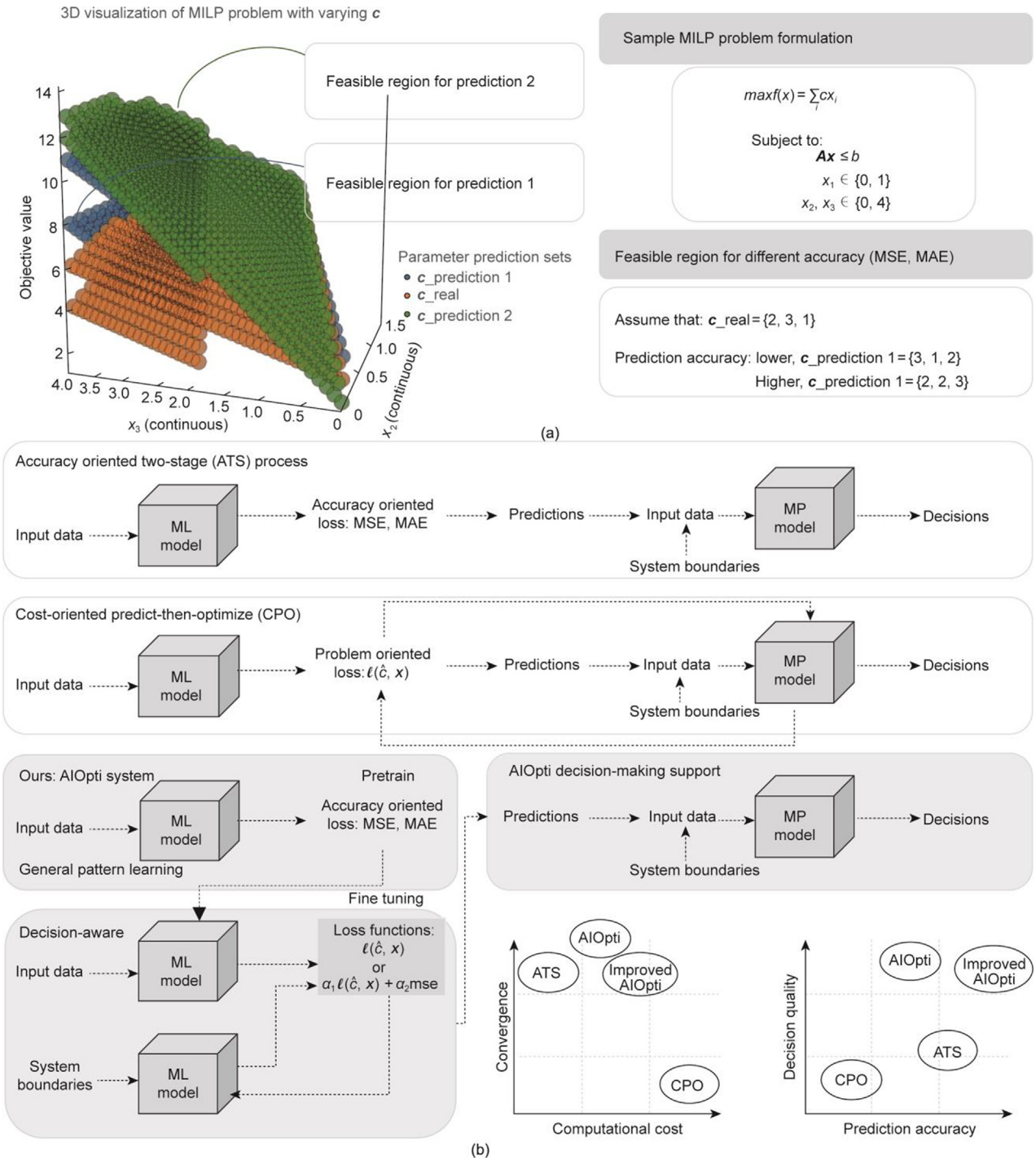


Fig. 1. Research motivation and proposed integration framework. (a) Visualization of the feasible region for different predictions with the same accuracy, based on a sample MILP model. The model, which includes three decision variables (one binary variable and two continuous variables), demonstrates how improving prediction accuracy can push the feasible region further away from the true feasible region. (b) Diagrams illustrating cooperation schemes between ML models and MP models, including the accuracy-oriented two-stage (ATS) process, the recently proposed cost-oriented predict-then-optimize (CPO) approach, and the proposed artificial intelligence (AI)-native optimization (AIOpti) system and improved_AIOpti frameworks. (c) Performance comparison of ATS, CPO, AIOpti, and the improved_AIOpti framework in addressing complex problems, evaluated in terms of computational cost, convergence, prediction accuracy, and decision quality. MSE: mean squared error; MAE: mean absolute error; $f(x)$: objective function; $\ell(\hat{\boldsymbol{c}}, x)$: problem-oriented loss functions; $\boldsymbol{A}$: constraint coefficient matrix; x: decision variable vector; b: constraint bounds; $\boldsymbol{c}$: objective coefficient vector; $\hat{\boldsymbol{c}}$: predicted objective coefficient vector; x_1–x_3: decision variables; α_1 and α_2: weight factors of loss functions.

analyses and misleading conclusions. Therefore, a salient question is whether it is possible to construct an artificial intelligence (AI)-native optimization (AIOpti) system that enables AI to be aware of the impact of predictions on decision-making.

To ensure ML predictions adhere to physical laws such as power-flow equations governed by partial differential equations (PDEs), recent studies have embedded these constraints directly into ML models as hard-coded formulations [17,18]. This method is particularly effective for low-level physical tasks such as state estimation, voltage stability analysis, and real-time grid simulation, where maintaining consistency with Kirchhoff's laws is critical for accuracy and system safety. However, when addressing

high-level energy system management tasks such as power scheduling, economic dispatch, and demand response optimization, the direct incorporation of PDE constraints becomes considerably less practical. These applications typically involve broader system-wide decision-making across extended time horizons and highly diverse operational conditions. Moreover, they often demand computational efficiency for real-time inference, flexibility to accommodate dynamic system topologies and user behaviors, and scalability to handle large-scale energy networks with significant uncertainty. In these contexts, embedding complex PDE structures can result in excessive computational overhead and introduce rigidity that limits the model's adaptability and generalizability.

To overcome these limitations, the cost-oriented predict-then-optimize (CPO) framework has gained prominence as a more flexible and scalable alternative. By decoupling prediction from optimization, CPO enables the use of efficient predictive models while preserving the ability to perform robust and goal-aligned decision-making through dedicated downstream optimization modules. Rather than optimizing for predictive accuracy alone, this framework guides the learning process to improve the practical value of decisions informed by predictions. As shown in Fig. 1(b), this end-to-end learning structure allows the model to account for optimization outcomes during training, ultimately increasing the real-world relevance and impact of the predictive component. There are two main aspects of CPO research: developing an optimally trained predictor using MP methods and investigating how to transfer cost information through gradient updating in the training process.

To train an optimally trained predictor, the first step is usually to reconstruct the empirical risk minimization (ERM) problem as an explicit mathematical model that integrates predictor training and task optimization in a closed-loop manner. In the second step, this ERM model will be solved to produce optimally trained predictors [19]. Bi-level programming-based ERM approaches have been further proposed to capture the sequential structure between prediction and optimization, training the predictor at the upper level while mimicking the optimization task at the lower level [15,19,20]. However, these methods often rely on limited training samples, which can result in overfitting. Additionally, they underestimate the impact of advanced forecasting technologies, and further validation is necessary to ensure effectiveness.

Using gradient updating to transfer cost information is challenging because many cost objective functions are piecewise constant, making it difficult to find effective first-order derivatives for the optimization problem. Two main methodologies are used to address this challenge: finding alternative gradient information to update the model parameters [21] and redesigning an alternative loss function with a non-zero gradient [22].

For finding alternative gradient information, the Karush–Kuhn–Tucker (KKT) condition is an attractive option because it makes optimization problems with hard constraints differentiable [22–25]. However, KKT-based methods require specific quadratic or conic programming solvers and necessitate additional objective terms for linear programming. Consequently, their efficiency and accuracy are limited compared with those of commercial MILP solvers [22].

Several methods have been proposed to design surrogate loss functions or gradients, including the smart "predict-then-optimiz e" plus (SPO+) loss function [16], differentiable black-box (DBB) solver [26], differentiable perturbed optimizer (DPO) [27], noise-contrastive estimate (NCE) [28], and learning to rank (LTR) [29]. These methods have fewer limitations regarding the type of optimization problem, making them more flexible for various applications. However, MP models often exhibit a complex solution space, such that accounting for the intricate interactions between predictions and this solution space can substantially increase the risk of ML model instability or collapse [30]. This complexity can complicate the convergence of the training process, making it more challenging to achieve stable and effective learning. Previous studies that have designed and validated surrogate loss functions have primarily focused on simple linear predictors, overlooking the challenges of convergence in more complex ML models. In the field of energy system management, practical problems such as price forecasting, load forecasting, and renewable power forecasting often require sophisticated ML models to achieve acceptable accuracy. Consequently, there is a growing need for further validation using advanced predictive technologies based on the CPO framework to address these complex engineering problems. Overall, research challenges remain in the following areas:

(1) How does the accuracy of ML models impact the performance of MP models in enhancing decision-making for energy system challenges?
(2) How can constrained optimization problems be effectively integrated into ML models?

To address these challenges, this paper introduces an AIOpti that significantly surpasses previous work by enabling AI to be aware of the impact of predictions on decision-making in an effective and flexible manner. This study demonstrates that directly using surrogate loss functions is inadequate for complex forecasting models coupled with sophisticated optimization tasks. Consequently, AIOpti enables ML models to understand the impact of their predictions on decision-making within MP models by employing a pretraining and fine-tuning scheme. This ensures that predictors initially achieve sufficient predictive accuracy, effectively approximating the optimized decision-making process. Subsequently, surrogate loss functions are employed to fine-tune the predictors further, increasing convergence stability and significantly reducing overall training time.

Specifically, we propose an innovative two-stage training framework: an initial accuracy-oriented pretraining phase, followed by a problem-oriented fine-tuning phase. During the pretraining phase, the focus is on learning general patterns and features from a large dataset to achieve robust prediction accuracy. This phase is highly adaptable, allowing for the integration of a range of AI techniques to optimize the model's performance. After the ML model achieves a foundational understanding of the general principles, the model transitions to the fine-tuning phase, in which the MP model is introduced using tailored surrogate loss functions. In the fine-tuning phase, the focus shifts toward problem-specific objectives, such as minimizing the operational costs, in order to strengthen the predictive model's capacity for guiding effective decision-making in real-world applications. By deeply fusing the ML and MP models, AIOpti bridges the gap between predictive accuracy and practical decision-making, ensuring that ML models not only predict effectively but also align with the cost-related priorities of complex systems. AIOpti represents a novel role for AI as a decision-aware assistant, supporting decision-making in complex scenarios.

We have validated our AIOpti system across various model components and two widely explored optimization problems. First, we developed a prediction model for day-ahead forecasting of electricity prices—a critical factor in promoting the integration of distributed energy sources and enabling demand-side response programs. Subsequently, we formulated optimization models targeting real-world energy system challenges. To improve AI's decision-making capabilities, we propose a deep fusion framework that effectively embeds optimization models within the predictive modeling process, thereby improving decision quality and making the models more applicable to real-world smart decision-making scenarios. Fig. 1(b) presents the specific structure of our AIOpti system. Moreover, we introduce an improved AIOpti system that not

only optimizes prediction accuracy but also improves decision quality, tailored to problem-specific needs. We have validated our AIOpti system on two prominent energy system challenges: virtual power plant (VPP) operation and distributed energy system management [31,32]. The experimental results revealed non-positive relationships between prediction accuracy and decision quality within optimization problems. Our AIOpti system demonstrated notable improvements in decision quality, with gains ranging from 1.10% to 18.81% compared with those of accuracy-oriented methods. Furthermore, compared with traditional problem-oriented approaches, our AIOpti system significantly increased decision quality (from 6.50% to 68.49%), prediction accuracy (from 33.63% to 83.67%), and training efficiency (from 38.99% to 87.03%). Overall, as shown in Fig. 1(c), the proposed AIOpti system advances the integration of AI and optimization models in solving complex engineering problems, promoting smarter and more efficient decision-making in energy systems.

2. Methods

2.1. Forecasting module

Data cleaning is essential for enhancing the performance of deep learning (DL) models, as raw datasets often contain noise and anomalies that negatively impact model accuracy and effectiveness. In this study, the raw dataset underwent a comprehensive cleaning process beginning with outlier detection and removal using the interquartile range (IQR) method. The IQR, a robust statistical measure of variability, identifies and eliminates extreme values effectively without being heavily influenced by outliers or skewed data distributions. Subsequently, an exponential moving average (EMA) was employed to smooth the data, mitigating short-term fluctuations and highlighting underlying trends. Additionally, Fourier filtering was applied as a further denoising step, removing high-frequency noise components while preserving the significant patterns inherent within the dataset.

Following the denoising of a dataset, feature engineering plays a critical role in improving predictive performance. In the context of electricity markets, readily available data typically includes electricity prices, forecasted load, actual load, and power generation. Leveraging domain knowledge, this study emphasizes the importance of the supply–demand balance, as the discrepancy between power generation and load is a key determinant of electricity price dynamics. Accordingly, an additional feature was constructed to represent the difference between power generation and load, aiming to reveal the intrinsic relationships among electricity prices, supply, and demand. To further address the issue of non-stationarity, which is prevalent in time-series data and may obscure true temporal patterns, we applied second-order differencing to transform non-stationary features into stationary ones. This transformation is essential for removing long-term trends and nonlinear drifts, thereby enabling DL models to focus on genuine fluctuations. The final model input integrates both differenced and original features, supplemented by time index features to capture the temporal structures inherent in electricity price signals.

In recent years, DL architectures have gained significant attention in time series forecasting due to their capacity to model complex temporal dependencies and nonlinear patterns in sequential data. To address the challenges associated with forecasting highly volatile electricity prices, we propose a hybrid model, ResNetLSTM, which combines the powerful feature-extraction capabilities of residual networks (ResNet) with the temporal modeling strengths of long short-term memory (LSTM) networks. This integration enables the model to effectively capture both spatial feature representations and long-term temporal dependencies, making it well-suited for high-resolution electricity price prediction tasks. As shown in Fig. 2, the ResNetLSTM model presents a hybrid DL architecture that integrates residual learning, sequence modeling, and attention-based fusion for time-series forecasting tasks. The model begins by flattening the input time-series data and extracting high-level global representations through a stack of fully connected residual blocks, thereby enabling deep feature propagation and mitigating vanishing gradient issues. These global features are subsequently reshaped into a sequential format and passed into an LSTM network, which captures temporal dependencies across the sequence. The LSTM's final hidden state is branched into two parallel prediction heads—one for the primary value forecast and the other for estimating variance. The outputs of these branches are concatenated and further refined through a learnable fusion mechanism enhanced by an attention module that adaptively weighs the contributions of value and variance. This modular architecture not only increases the predictive accuracy by combining spatial and temporal learning but also improves the interpretability and robustness through dual-branch uncertainty modeling and attentive integration.

2.2. Optimization module for VPPs

The decision module is built upon optimization models. The constraints and objectives of models can be various, depending on the targeted practical problem. Here, we take the day-ahead control of energy-storage-based VPPs as the targeted problem, because it is a widely investigated energy system management problem that is significantly influenced by electricity prices. VPPs are digital twin systems that integrate various distributed energy resources (DERs) such as solar panels, wind turbines, batteries, and flexible demand units, all working together as a single power plant to participate in the electricity market.

Comprehensive system control of VPPs can be computationally consuming, so the models are usually developed in a mixed-integer programming (MIP) format. It takes several to hundreds of seconds to solve such a model once. Consequently, the training time of DL models in a problem-oriented DL framework will be

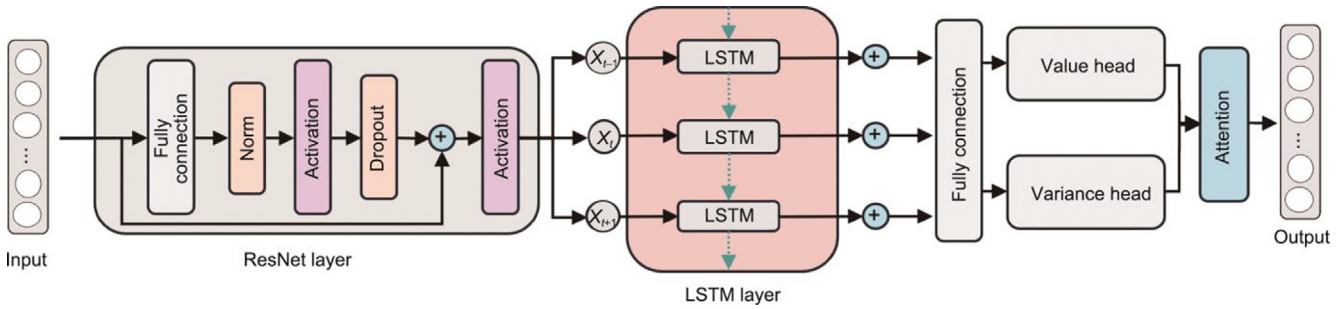


Fig. 2. The specific model architecture of the proposed ResNetLSTM. *X*: input features; *t*: time steps.

extremely long: The training dataset is large, and the number of times the optimization model must be solved in every epoch is equal to the number of data samples. Therefore, the complex optimization model is reconstructed into two-staged sub models.

A first-stage sub-model (M1) is established to optimize the initial schedule plans of the energy hub, including the day-ahead operation plans of wind turbines, gas turbines, solar photovoltaics, energy-storage systems, and demand-response programs. The demand-response program includes 20 households. A detailed mathematical model is provided in Appendix A. The M1 model provides the necessary system boundaries for the further operational management of energy-storage-based VPPs. Based on the outputs of M1, the second-stage sub-model (M2) is developed to optimize the interactions between the energy-storage-based VPPs and the power grid. To increase computational efficiency, the electric vehicles (EVs) and home battery storage systems of 20 households in region M2 are aggregated into a single integrated EV system and a centralized battery system.

M2 aims at maximizing profit based on forecasted electricity prices. The objective function f can be expressed as follows:

$$\max f = \sum_t c_t^{\text{elec}} \left(\text{PSE}_t - P_t^{\text{grid}} \right) \tag{1}$$

$$\text{pse}^{\min} \text{BPS}_t \leq \text{PSE}_t \leq \text{pse}^{\max} \text{BPS}_t, \forall t \in T \tag{2}$$

wherein c_t^{elec} represents the forecasted electricity in time interval t, P_t^{grid} denotes the power purchased from the main grid, and PSE_t refers to the electricity transaction behaviors of VPPs. $\text{PSE}_t > 0$ means that electricity is sold to the power grid. $\text{pse}^{\min}$ and $\text{pse}^{\max}$ denote the minimum and maximum power sold to the grid, and BPS_t is a binary variable indicating whether power is sold. T represents the set of time intervals.

M2 had three main types of constraints: system balance constraints, energy storage batteries constraints, and EV operational constraints.

Balancing the energy flows is the primary requirement for the energy system, and this constraint is expressed as Eq. (3). Parameters with the symbol "*" are outputs from M1, including the outputs of wind turbines $P_t^{\text{w},*}$, solar photovoltaics $P_t^{\text{s},*}$, gas turbines $P_t^{\text{g-p},*}$, and the load profile $L_t^{\text{total},*}$ after scheduling. For the energy-storage-facilities-based VPP, electricity-transaction behaviors and operations containing the charging and discharging of batteries ($P_t^{\text{bst}}, \text{PD}_t^{\text{bst}}$) and EVs ($P_t^{\text{V2H}}, \text{PD}_t^{\text{V2H}}$) need to be rescheduled.

$$P_t^{\text{grid}} + P_t^{\text{w},*} + P_t^{\text{s},*} + P_t^{\text{g-p},*} - \text{PSE}_t = L_t^{\text{total},*} + P_t^{\text{V2H}} - \text{PD}_t^{\text{V2H}} + P_t^{\text{bst}} - \text{PD}_t^{\text{bst}}, \forall t \in T \tag{3}$$

Detailed formulations of the energy-storage system and vehicle-to-grid services are provided in Appendix A.

2.3. Optimization module for the distributed energy system

Eqs. (1) and (2) can also be used to present the objective of a distributed energy system in this context, which is to actively participate in power transactions within the electricity market, leveraging dynamic electricity pricing to optimize arbitrage opportunities. In this framework, the generation cost of DERs is often disregarded, as the primary focus is on maximizing the utilization of renewable energy sources. This approach not only aligns with reducing overall energy costs but also supports the transition toward sustainable energy systems by prioritizing the integration of clean, renewable power.

Eq. (4) ensures the balance between power supply and demand, where P_t^{pv} denotes the power generated by photovoltaic (PV) systems.

$$P_t^{\text{grid}} + P_t^{\text{w}} + P_t^{\text{g-p}} + P_t^{\text{pv}} - \text{PSE}_t = L_t^{\text{total}} + P_t^{\text{bst}} - \text{PD}_t^{\text{bst}}, \forall t \in T \tag{4}$$

The remaining constraints, including the operation of the distributed generators and the battery energy-storage system, are provided in Appendix A.

2.4. AIOpti system

This paper aims to equip AI with an awareness of how its predictions influence decision-making. Previous studies have typically separated the training processes of accuracy-oriented ML models and cost-oriented ML models, comparing their regret on the test dataset in order to validate the advantages of CPO models. However, this method overlooks the fact that cost-oriented predictions are fundamentally based on achieving acceptable prediction accuracy. Furthermore, the use of large training datasets, complex ML models, and optimization frameworks presents significant challenges for convergence and acceptable training times.

To address this issue, the proposed approach enables ML models to understand the impact of their predictions on decision-making within MP models by employing a pretraining and fine-tuning scheme. We propose a novel training framework that first employs an accuracy-oriented pretraining phase, followed by a problem-oriented fine-tuning phase. During the pretraining phase, the focus is on achieving acceptable accuracy by enabling the ML model to learn general patterns and features from a large dataset. This process is flexible, allowing for the integration of various AI technologies to optimize model performance.

Xu et al. [30] demonstrated the effectiveness of incorporating prior knowledge into DL models by combining a rule-based loss function during the fine-tuning process. Researchers have designed various surrogate loss functions or gradients to extract gradient information from the objective functions of optimization problems. To leverage the advantages of these methods, this paper introduces optimization problems into ML models by employing different surrogate loss functions in the fine-tuning process.

Five different surrogate loss functions or gradients are briefly introduced, as follows:

SPO+ [16] is a convex and sub-differentiable surrogate loss function specifically designed for problems with linear objective functions.

$$\ell_{\text{SPO+}}(\hat{c}, c) = \max_{w \in V} \{ c^{\text{T}} w - 2\hat{c}^{\text{T}} w \} + 2\hat{c}^{\text{T}} w^*(c) - z^*(c) \tag{5}$$

wherein $\hat{c}$ represents the predicted value of the cost vector, z^* refers to the optimal objective value, and V denotes the set of feasible solutions. The decision variables in the objective function are referred to as w, indicating the optimal objective value of the optimization model.

The DBB [26] method computes a subgradient through the continuous interpolation of solutions. While it is constrained by linear objective functions, it offers flexibility in the choice of the loss function. The loss gradient can be calculated as follows:

$$\nabla f_\lambda(c) = -\frac{1}{\lambda} [w(c) - w_\lambda(c)] \tag{6}$$

where $\nabla f_\lambda(c)$ indicates the loss gradient, and the hyper-parameter $\lambda > 0$ controls the tradeoff between the informativeness of the gradient for effective optimization and the faithfulness to the original function. A larger λ prioritizes gradient informativeness, while a smaller λ ensures the model stays closer to the original function.

Based on DPO, random noise can be introduced into the coefficients of the objective function to probabilistically smooth the

piecewise constant solutions. Using the duality principle, the Fenchel–Young (PFY) loss is then constructed [27]:

$$\ell_\varepsilon(c,w) = F_\varepsilon(c) + \varepsilon\Omega(w) - \langle c, w\rangle \tag{7}$$

where $F_\varepsilon(c)$ is the perturbed objective function, ε is a positive perturbation parameter, $\Omega(w)$ is a regularization function depending on w, and $\langle c, w\rangle$ represents the dot product between c and w.

The NCE loss formulation is designed to distinguish between the probability of the optimal solution and the negative samples (or "noise") [28], with various adaptations also presented in the literature. The NCE based maximum a posteriori (MAP) loss function is defined by Eq. (8b). Here, $\widehat{c}$ can be replaced by $\widehat{c} - c$ for linear objective functions to improve performance.

$$\ell_{\text{NCE}} = \sum_{i\in I}\sum_{w^s\in S}\left(f\left(w_i^*,\widehat{c}\right) - f\left(w_i^s,\widehat{c}\right)\right) \tag{8a}$$

$$\ell_{\text{NCE_MAP}} = \sum_{i\in I}\sum_{w^s\in S}\left(f\left(w_i^*,\widehat{c}\right) - f\left(w_i^{s,*},\widehat{c}\right)\right) \tag{8b}$$

wherein $f(w,c)$ is the objective function, I indicates the training dataset, and S is a set of negative samples w^s, which are non-optimal or "noise" solutions. $w_i^{s,*}$ is the MAP solution for the current model.

Based on NCE, LTR [29] transforms the predict-then-optimize task into a ranking problem, where it ranks the subset of feasible solutions according to their objective values. $\ell_{\text{LTR}}^{\text{PR}}$ denotes the pointwise ranking loss function, $\ell_{\text{LTR}}^{\text{Pair}}$ refers to the pairwise loss function, and $\ell_{\text{LTR}}^{\text{List}}$ represents the listwise loss function.

$$\ell_{\text{LTR}}^{\text{PR}} = \frac{1}{|S|}\sum_{w\in S}\left(f\left(w,\widehat{c}\right) - f(w,c)\right)^2 \tag{9}$$

$$\ell_{\text{LTR}}^{\text{Pair}} = \frac{1}{\left|\text{OP}_c^S\right|}\sum_{(p,q)\in \text{OP}_c^S}\max\left(0, v + f\left(w^p,\widehat{c}\right) - f\left(w^q,\widehat{c}\right)\right) \tag{10}$$

$$\ell_{\text{LTR}}^{\text{List}} = -\frac{1}{|S|}\sum_{w\in S} p_\tau(w|c)\log_{p_\tau}\left(w|\widehat{c}\right) \tag{11}$$

wherein (p,q) is an ordered pair in S with respect to c. OP_c^S is the set of all ordered pairs where the first coordinate dominates the second in the order induced by c. $v > 0$ is the margin value. Let τ be the temperature parameter that controls the smoothness of the distribution, and p_τ represents the cross-entropy loss of these probabilities. LTR provides the flexibility to handle various types of optimization problems, which is a significant advantage. However, there is a lack of experimental validation for nonlinear problems.

In Algorithm 1, AIOpti achieves deep integration of ML and MP through a three-phase pipeline: it begins with problem formulation, in which the feature vector $\boldsymbol{x}$, forecasting target $\boldsymbol{c}$, and operational constraints $\boldsymbol{Ax} = b$ are defined and used to construct both ML and MP models tailored to the decision task. Then, it proceeds to an accuracy-oriented pretraining phase, during which the ML model is trained on large-scale data to capture fundamental input–output relationships and maximize predictive performance using advanced AI techniques. Finally, it enters a decision-oriented fine-tuning phase, in which the MP model is coupled to the ML model via a customized surrogate loss that embeds problem-specific objectives (e.g., cost minimization, resource-allocation efficiency), thereby aligning predictions directly with operational goals and ensuring actionable, cost-effective solutions.

Algorithm 1 (*Fusion mechanism of ML and MP models*).

Input: $\boldsymbol{A}, b$; training data including feature vectors $\boldsymbol{x}$ and forecasting objective $\boldsymbol{c}$
Hyperparameters: learning rate α, epochs, batch size

Initialize ω
for each epoch **do**
  for each batch **do**
    $\widehat{c} \leftarrow \text{ML}(\omega, x)$
    loss $\leftarrow \text{MSE}(\widehat{c}, c)$ or $\text{MAE}(\widehat{c}, c)$
    $\nabla\omega \leftarrow \frac{\partial \text{loss}}{\partial \omega}$
    $\omega = \omega - \alpha\nabla\omega$
  end for
end for
Construct problem-oriented training data: $x, c, w^*(c), z(c) \leftarrow \text{MP}(Ab, c)$
Load ω^*
for each epoch **do**
  for each batch **do**
    $\widehat{c} \leftarrow \text{ML}(\omega^*, x)$
    $w^*(\widehat{c}), z(\widehat{c}) \leftarrow \text{MP}(Ab, \widehat{c})$
    loss $\leftarrow \text{CPO loss}\left(\left[\widehat{c}, w^*(\widehat{c}), z(\widehat{c})\right], [c, w^*(c), z]\right)$
    $\nabla\omega^* \leftarrow \frac{\partial \text{loss}}{\partial w^*(\widehat{c})}\frac{\partial w^*(\widehat{c})}{\partial \widehat{c}}\frac{\partial \widehat{c}}{\partial \omega}$
    $\omega^* = \omega^* - \alpha\nabla\omega^*$
  end for
end for

MSE: mean squared error; MAE: mean absolute error.

2.5. Automatic tuning for hybrid loss

Sang et al. [33] demonstrated the benefits of using hybrid loss to improve model performance. However, the hyperparameters—specifically the weights of the MSE loss and SPO+ loss—were manually adjusted and tested only a few times. To address this limitation, we propose an AIOpti-based Bayesian optimization framework to optimize these parameters, further improving the model's performance in terms of both accuracy and its ability to guide the operation of urban energy systems.

Bayesian optimization is a strategy for optimizing black-box functions that are expensive to evaluate. It works by creating a probabilistic model of the objective function, often using Gaussian processes (GPs), to guide the search for the optimal solution. This approach balances exploration (searching unexplored regions) and exploitation (refining known promising areas) by updating the model with past evaluations and selecting the next point based on an acquisition function. Due to its high efficiency, Bayesian optimization is widely applied in areas such as hyperparameter tuning and control and decision-making systems, making it well-suited for the automatic tuning of hybrid loss functions. The hybrid loss function ℓ_{hybrid} is defined in Eqs. (12) and (13). ℓ_{accuracy} represents the accuracy-oriented loss function, and ℓ_{cost} denotes the cost-oriented loss function. Bayesian optimization is used here to optimize the weight factors of the different purpose-oriented loss functions in order to minimize regret. Regret [34] quantifies the error associated with not choosing the best possible option; it can be calculated using Eq. (14) for a maximization problem:

$$\ell_{\text{hybrid}} = \alpha_1\ell_{\text{accuracy}} + \alpha_2\ell_{\text{cost}} \tag{12}$$

$$\alpha_1 + \alpha_2 = 1 \tag{13}$$

$$\text{Regret} = c^{\mathrm{T}} w(c) - c^{\mathrm{T}} w^{*}(\hat{c}) \tag{14}$$

3. Results

The data used in this paper is sourced from the transparency platform ENTSOE—more specifically, from the Germany–Luxembourg (DE-LU) day-ahead electricity market [35]. It includes 22 distinct types of power-generation data. Additionally, the dataset contains information on day-ahead total load forecast, actual total load, and day-ahead price. All data are recorded at 15 min intervals for two years (2019 and 2020).

3.1. Non-positive relationship between prediction accuracy and decision quality

Firstly, we present quantitative evidence demonstrating that improved ML prediction accuracy does not always translate into enhanced decision quality and may even result in poorer decision outcomes in certain cases. A range of ML models commonly used for predictive tasks were evaluated; detailed descriptions of the model configurations are provided in Section 2 and the Appendix A. In this study, the models were applied to forecast electricity prices 96 time-steps ahead, using historical data from the previous one and a half days. The performance of the models was assessed along two key dimensions: prediction accuracy from a computational perspective and decision quality from an engineering standpoint.

As presented in Fig. 3(a), while the primary objective of this study is not to push the boundaries of ML models for electricity price forecasting, our ResLSTM model exhibits superior predictive performance, with a mean squared error (MSE) of 122.91, mean absolute error (MAE) of 8.30, symmetric mean absolute percentage error (SMAPE) of 25.60%, and median absolute error (MedAE) of 6.28. Nevertheless, from an engineering perspective, higher predictive accuracy does not necessarily equate to better decision-making. For example, when comparing the performance of the temporal convolutional network (TCN) and LSTM models, while the MAE was reduced from 13.99 to 11.64 (+20.19%), the regret increased markedly from 12.40 to 14.26 (−15.00%). The regret [34] quantifies the error associated with not choosing the optimal possible option. Similarly, in cases where the MAE improved from 9.35 to 8.75 (+6.86%), the regret paradoxically worsened, rising from 7.65 to 7.70.

The mean expected benefits (MEBs) are the optimal benefits derived from the predicted values. However, relying solely on MEB can be misleading for decision-makers, as it may result in either overestimation or underestimation of the actual benefits. Notably, the nonlinear relationship between MEB and decision quality, as illustrated in Figs. 3(a) and (b), indicates that improved predictive accuracy does not necessarily translate into superior decision outcomes. This disconnect underscores the potential for increased decision-making losses, even in the presence of high model performance. Fig. 3(c) illustrates the nonlinear relationship between accuracy metrics and decision quality, highlighting the necessity for more tailored ML models to better guide applications in complex real-world scenarios.

3.2. AIOpti empowers smart and effective decision-making in VPPs

As we have demonstrated, improving prediction accuracy does not consistently increase decision quality. The next challenge addressed in this paper is how to efficiently train a predictive ML model that provides robust guidance for optimization models to support better decision-making. Although prior studies [16,26–29] have introduced methods to integrate predictive ML and optimization models, our experiments reveal the limitations of these approaches when applied to complex ML models and engineering problems, underscoring the need for more adaptable methodologies.

In contrast to the traditional two-stage process, where an accuracy-oriented predictive model is trained first, and its predicted values are subsequently used as inputs for optimization models, previous CPO approaches have focused on deriving surrogate gradient information. However, they largely underestimate the convergence difficulties posed by complex prediction and decision-making tasks, particularly when applied to intricate engineering problems. Fig. 4(a) quantifies the limitations of these CPO approaches. Six training methods—including SPO+, DBB, PFY, NCE, NCE_diff (using $\hat{c} - c$), and LTR—are tested. Most perform worse than the proposed method in terms of both accuracy and decision quality.

Compared with the two-stage process approach based on ResLSTM, the regrets for DBB, PFY, NCE_MAP, and LTR increased significantly from 6.38 to 8.64 (+35.42%), 7.45 (+16.77%), 14.00 (+119.44%), and 9.24 (+44.83%), respectively. Only SPO+ and NCE_diff showed improvements in regret of 12.23% and 9.72%, respectively. In terms of accuracy, these approaches exhibited highly unacceptable performance, with MAE values reaching as high as 98.08. These experimental results underscore the inadequacy of existing integration methods when applied to the intersection of complex prediction tasks and intricate optimization problems. Thus, there is a critical need to develop more advanced, generalizable, and efficient approaches to better align predictive ML models with optimization processes.

To address the limitations of capability and inefficiency, we demonstrate the superior performance of the proposed integration framework in increasing the decision quality, transferability, and training efficiency. As illustrated in Fig. 4(a), the application of the proposed AIOpti system significantly improved both accuracy and decision quality. For instance, with respect to the MAE metric, SPO+ decreased from 21.15 to 8.83, representing a substantial reduction of 58.25%. Similarly, DBB dropped from 46.97 to 9.66 (−79.43%), PFY decreased from 41.15 to 18.84 (−54.22%), NCE_MAP was reduced from 98.08 to 16.02 (−83.67%), NCE_diff decreased from 22.54 to 14.96 (−33.63%), and LTR decreased from 17.84 to 8.42 (−52.80%). In terms of the decision quality metric, there was also a substantial reduction in regret. Specifically, SPO+ decreased from 5.60 to 5.18, representing a 7.50% reduction. In addition, DBB dropped from 8.64 to 5.64 (−34.72%), PFY from 7.45 to 5.68 (−23.76%), NCE_MAP from 14.00 to 6.14 (−56.14%), NCE_diff from 5.76 to 5.40 (−6.25%), and LTR from 9.24 to 6.31 (−31.71%). These reductions highlight the framework's capacity to significantly increase decision-making quality by minimizing regret. These findings underscore the remarkable ability of the AIOpti system to enhance model convergence and overall performance.

Compared with accuracy-oriented models, the AIOpti system enables ML models to achieve superior decision outcomes. The experimental results indicate that the SPO+-based AIOpti system demonstrated the greatest improvement in decision quality, achieving an enhancement of 18.81%. Other CPO-based AIOpti approaches also exhibited notable improvements in decision quality: DBB showed a 11.60% improvement, PFY improved by 10.97%, NCE_MAP by 3.76%, NCE_diff by 15.36%, and LTR by 1.10%. These results underscore the effectiveness of the AIOpti system in driving better decision-making across various optimization approaches.

We further demonstrate the advantages of the AIOpti system in improving model convergence by examining the reduction in training time. A significant decrease in training time was observed

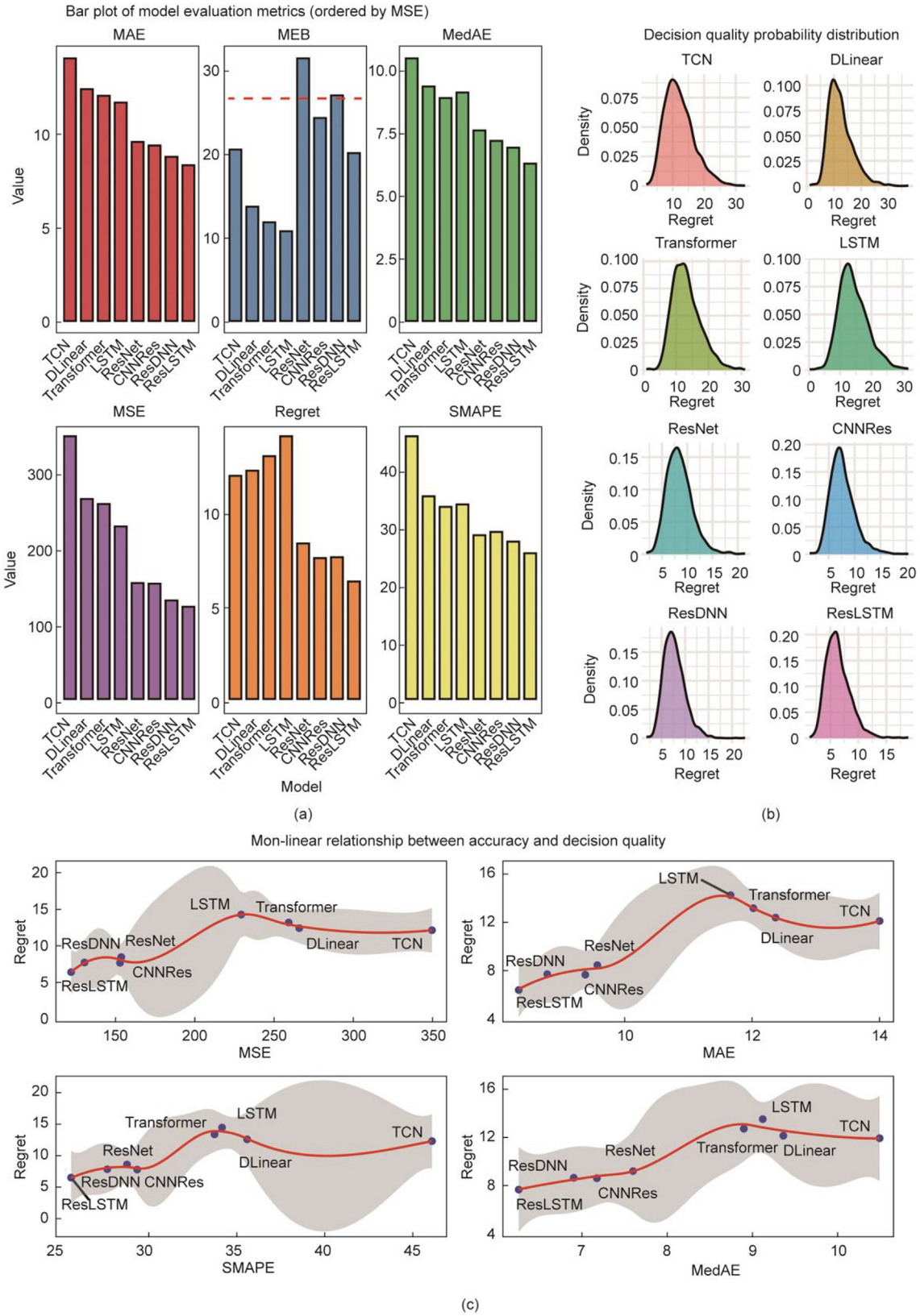


Fig. 3. Demonstration of the nonlinear relationship between accuracy metrics and decision quality. (a) Performance evaluation metrics for model accuracy, presenting various metrics assessing the accuracy of the tested models (MAE, MSE, MedAE, and SMAPE). The red dotted line shows the MAB, representing the mean achievable benefits based on real data. (b) Regret distribution across prediction models, calculated based on predicted values from different models. Lower regret values indicate that the decision based on the predictions is closer to the optimal solution, highlighting better model performance. (c) Relationships between accuracy and decision quality metrics, exploring the correlation between accuracy evaluation metrics (MAE, MSE, MedAE, and SMAPE) and decision quality metrics. This figure investigates how variations in accuracy measures impact the quality of decisions made using the models, providing insights into the tradeoffs between prediction precision and decision effectiveness.

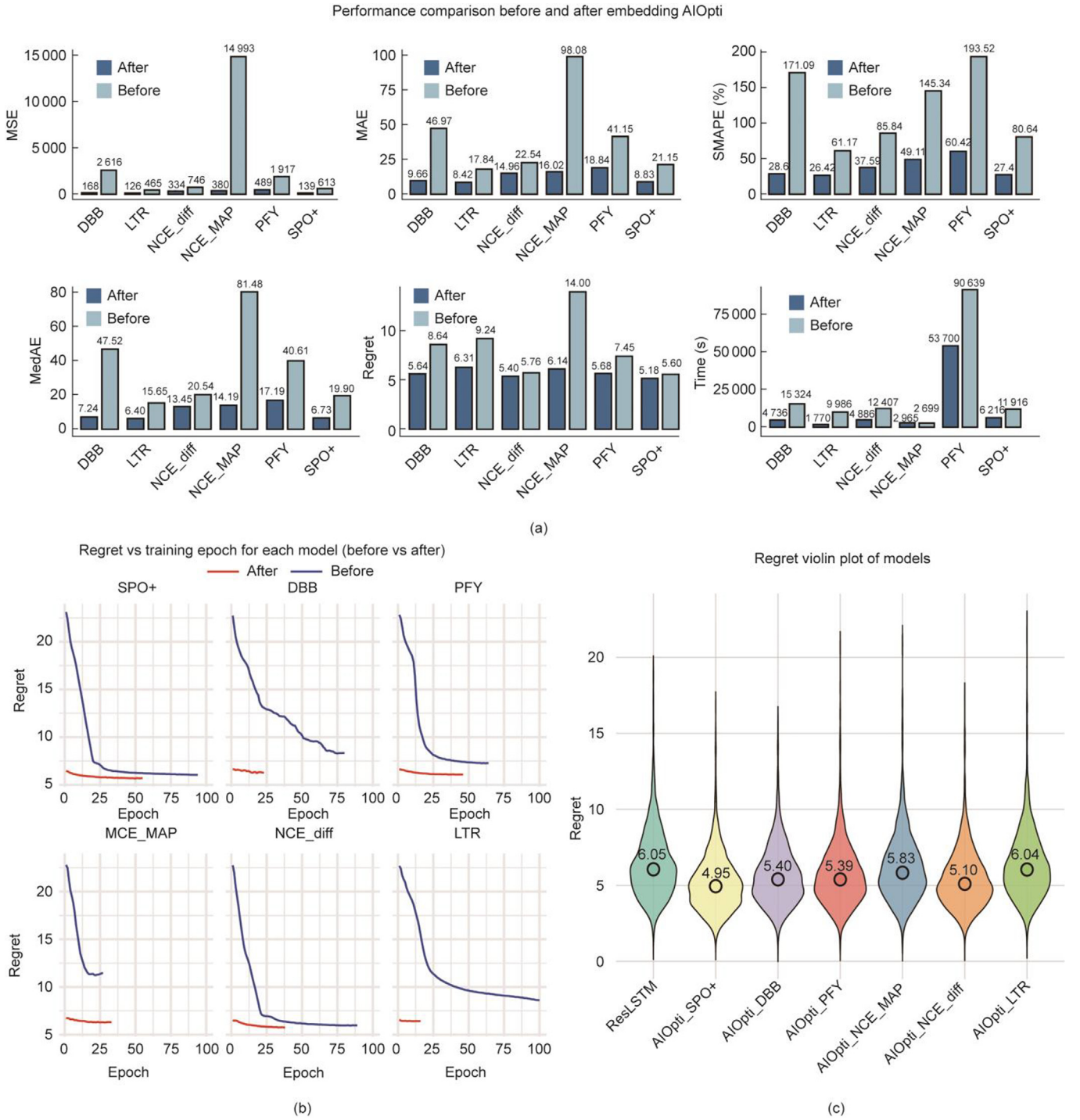


Fig. 4. Comparison of traditional problem-oriented training and AIOpti-embedded problem-oriented training. (a) Performance evaluation of models trained using traditional problem-oriented methods vs those incorporating AIOpti, across metrics including accuracy, decision quality, and training time. (b) Regret variation across training epochs; the blue line represents models without AIOpti embedding, and the red line indicates models after AIOpti embedding. (c) Regret distribution for AIOpti-embedded models compared with the accuracy-oriented ResLSTM model, demonstrating the AIOpti framework's effectiveness in improving decision quality. The dotted line marks the peak distribution density and the maximum regret values for each model.

after embedding the AIOpti system into the models. Specifically, the SPO+ training time decreased from 11 916 s to 6216 s, a reduction of 47.83%. Similarly, DBB, PFY, NCE_diff, and LTR showed reductions of 69.09%, 40.75%, 60.62%, and 82.28%, respectively. As illustrated in Fig. 4(b), the integration of the AIOpti system significantly improved model performance and accelerated convergence, reducing the number of training epochs required. This reduction in epochs directly contributes to decreased training time, as problem-oriented training necessitates solving the corresponding MP model repeatedly—for every sample and every epoch. Given that solving MP models is computationally intensive and primarily CPU-based, minimizing the number of epochs is crucial in improving overall computational efficiency. The experimental results validate the effectiveness of AIOpti in achieving faster convergence and

superior performance, thereby demonstrating its computational advantages. Fig. 4(c) illustrates the regret distribution, emphasizing the improved decision quality that resulted from the incorporation of the AIOpti system. These results highlight the substantial efficiency gains achievable by applying the AIOpti system, leading to faster training and improved model convergence.

3.3. Scalability of the AIOpti system

The CPO models were initially trained using a dataset of 7447 samples. To assess the performance of models with larger datasets, we conducted comparative experiments using a more extensive dataset of 22 339 samples. To optimize computational efficiency, we focused solely on the SPO+ approach in our comparative experiments.

Without incorporating the AIOpti system, enlarging the dataset significantly improved the model's predictive accuracy and decision-making performance. As demonstrated in Table 1, this larger dataset notably reduced decision regret from 5.60 to 5.17 and decreased the MAE from 21.15 to 10.67. However, these improvements came at the cost of substantially increased computational demands, with the training time escalating to 32 995 s, approximately three times longer than training with the smaller dataset, primarily due to the repeated optimization processes conducted in each epoch.

Integrating the AIOpti system into the training process further advanced the predictive performance, lowering the MAE to 8.72 and reducing the regret from 5.18 to 5.06. Importantly, AIOpti implementation exhibited only a modest increase in training duration. When compared directly with the standard SPO+ method, the AIOpti_SPO+ approach achieved a 61.17% reduction in computational time, underscoring AIOpti's substantial capability to increase computational efficiency in complex scenarios. Thus, embedding the AIOpti system effectively increases both predictive accuracy and decision quality, while notably accelerating model convergence.

To enhance model performance in both accuracy and decision quality, we introduce a hybrid loss function that integrates a weighted SPO+ loss with the MSE loss. Fig. 5(a) presents the weight factor, which is optimized through Bayesian optimization. The experimental results reveal substantial improvements with the refined AIOpti system, achieving optimal performance in both accuracy and decision quality. Fig. 5(b) demonstrates the exceptional performance of the improved_AIOpti system in capturing the dynamic trends of day-ahead prices, contributing to its lower regret, as illustrated in Fig. 5(c). Additionally, the figure highlights the model's capability to mitigate extreme scenarios. However, the Bayesian optimization process notably increased the training time, as it relies on iterative training to identify the optimized weight factor. This finding highlights the tradeoff between achieving superior performance and lowering the computational cost.

3.4. AIOpti empowers smart management of distributed energy systems

We further applied the proposed AIOpti system to the operational optimization of a distributed energy system. Fig. 6(a) demonstrates the limitations of conventional CPO approaches, which failed to train a cost-oriented predictive ML model with satisfactory performance. In contrast, the AIOpti system achieved enhanced decision quality, demonstrating up to an 18.60% improvement over accuracy-focused predict-then-optimize methods, despite exhibiting lower prediction accuracy. Compared with the CPO approaches, the AIOpti system showed significant improvements in both decision quality and prediction accuracy—up to 68.49% and 93.63%, respectively. Additionally, AIOpti achieved up to an 87.03% reduction in training time. Fig. 6(b) illustrates that the AIOpti system improves the convergence performance of model training. Furthermore, the AIOpti system enhances model performance in decision quality. Notably, Fig. 6(c) underscores that embedding the AIOpti system not only significantly enhances decision quality but also improves the lower bounds in worst-case scenarios, highlighting the AIOpti system's ability to provide robust decision-making guidelines even under uncertain conditions.

4. Conclusions

In this work, we analyzed the phenomenon of the non-transitive nature of the optimal state of ML models, which poses challenges for downstream decision-making in MP models. To address this issue, we presented a robust and generalizable AIOpti system that deeply fuses predictive ML models with complex MP models, offering a significant advancement in decision-making quality. Our AIOpti system excels even when applied to advanced ML models and intricate optimization scenarios, surpassing previous methodologies focused solely on accuracy- or problem-specific approaches. This deep fusion markedly improves decision outcomes, providing a substantial improvement in fields such as demand-response programs and distributed energy systems. The novelty of our approach lies in its ability to bridge the gap between predictive ML models and real-world applications. Traditionally, the limitations of ML models in practical decision-making are encapsulated by the aphorism that "all models are wrong, but some are useful." Our AIOpti system tackles this challenge by equipping AI with an awareness of how its predictions influence decision-making, thereby providing reliable decision support in engineering domains where uncertainty and complexity prevail.

Moving forward, a promising research direction lies in the deeper integration of multi-parameter prediction with nonlinear optimization. This approach has the potential to significantly expand the capability of AIOpti in delivering intelligent, high-quality decision support across a wider spectrum of real-world applications. Another valuable direction involves incorporating uncertainty information such as predictive distributions or confidence intervals into the framework. This would support more effective uncertainty modeling and contribute to the robustness of decision-making processes. Together, these efforts will further advance the adoption of data-driven, intelligent decision-making in engineering and other domains, unlocking new potential in environments characterized by dynamic and complex challenges.

Table 1
Performance of the improved_AIOpti model, large-dataset-trained SPO+ model, and AIOpti_SPO+ model.

Model	MSE	MAE	SMAPE	MedAE	Regret	MEB	Training time (s)
SPO+_BD	195.22	10.67	33.82	8.47	5.17	23.08	32 995
AIOpti_SPO+_BD	134.82	8.72	26.68	6.66	**5.06**	24.64	12 811
Improved_AIOpti	**132.05**	**8.64**	**26.41**	**6.62**	5.08	24.67	80 678

Bold values indicate the best performance.

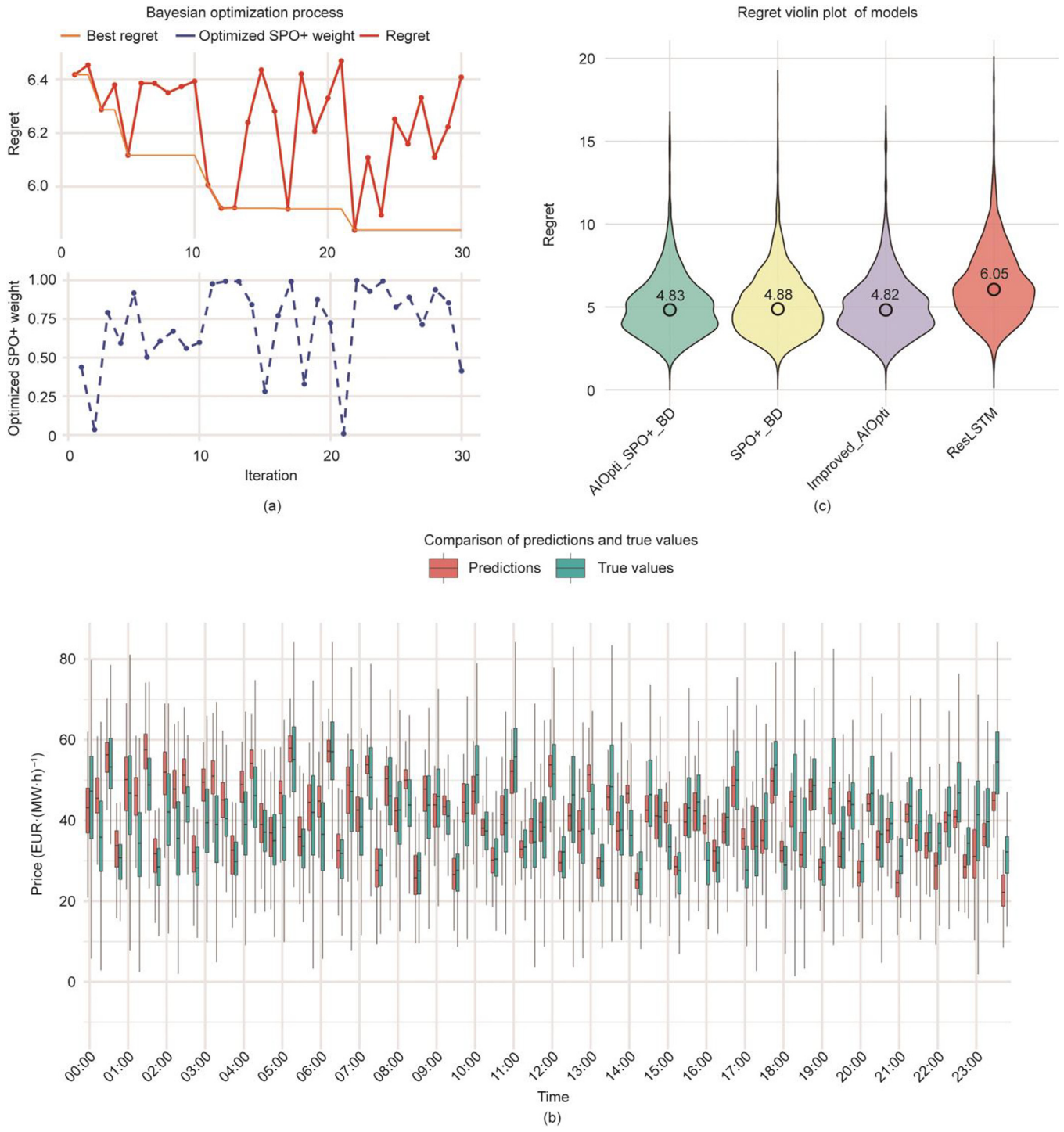


Fig. 5. Flexibility of the AIOpti system demonstrated using Bayesian optimization and large datasets. (a) Bayesian optimization process for identifying the optimal weight factor ($\ell_{\text{hybrid}} = (1-\alpha)\ell_{\text{mse}} + \alpha\ell_{\text{spo+}}, \alpha \in [0,1]$) between SPO+ loss and MSE loss. The red line represents the minimum regret for each iteration, the yellow line indicates the overall best regret across the optimization process, and the blue line shows the optimized SPO+ loss weight factor. (b) Comparison of the predicted values from the improved_AIOpti model against actual values over 96 time intervals. (c) Regret distribution for the improved_AIOpti model, the large-dataset-trained SPO+ model, and the AIOpti_SPO+ model. The numbers indicate the median regret values for each model.

Data availability

For the optimization models of VPPs and distributed energy system, basic data and parameters can be found in our previous paper [36].

Code availability

The code used in this research can be found at Github (https://github.com/logan6wgt/AIOpti).

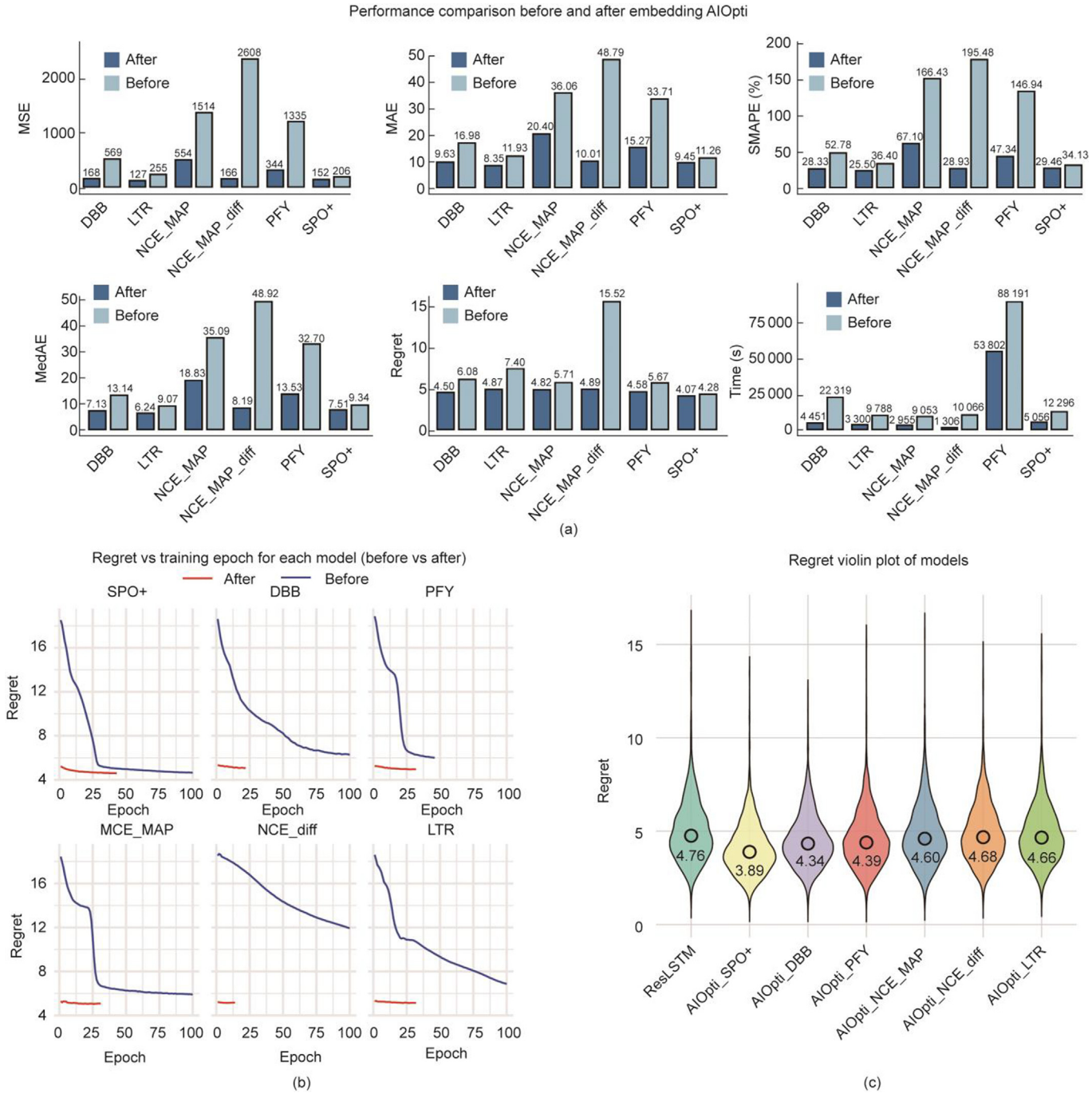


Fig. 6. Application of the AIOpti system in distributed energy system optimization. (a) Comparison of model performance before and after AIOpti-system embedding, showing significant improvements across accuracy and decision quality metrics. (b) The training process of various models, before and after embedding the AIOpti system, illustrating the superior performance of the AIOpti system in enhancing convergence. (c) Regret distribution for all models in distributed energy system optimization, highlighting the superior performance of the AIOpti-embedded model in improving decision quality and mitigating extreme scenarios.

CRediT authorship contribution statement

Guotao Wang: Writing – original draft, Conceptualization, Methodology, Investigation. **Zhenjia Lin:** Writing – review & editing, Investigation, Conceptualization. **Yuntian Chen:** Supervision, Investigation, Writing – review & editing, Conceptualization. **Haoran Ji:** Writing – review & editing, Conceptualization. **Dayin Chen:** Visualization, Data curation. **Haoran Zhang:** Supervision, Writing – review & editing. **Peng Li:** Writing – review & editing, Supervision. **Jinyue Yan:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eng.2025.06.037.

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